

EXACT HISTORY DYNAMICS, DOMAIN-SAFE MEMORY, AND UNIVERSAL EXCESS PATH PRESSURE

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ABSTRACT. Prepared Sinai paths are conditioned on a genuine past-only natural-extension history. The resulting future kernel is a weighted Feller semigroup and supports a correctly centered martingale-coboundary decomposition and quenched enhanced invariance principle. Nonlinear history pressure is normalized by the positive Feynman–Kac eigenfunction, and exact resolved memory is obtained from the full unresolved-space compressed resolvent with the causal Volterra sign.

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1. ROUND-TEN RECONSTRUCTION: A PAST MARKOV KERNEL, AN EIGENFUNCTION DOOB TRANSFORM, AND EXACT MEMORY

Let H be the weighted natural-extension past space. A point of H contains only coordinates at times $s \leq 0$; it does not contain the unstable/future itinerary. Disintegration of the prepared invariant path law over this past sigma-field gives a regular conditional future kernel $P_t(h, dh')$.

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1.1. The past Feller process.

Theorem 1.1 (Weighted past-kernel Feller semigroup). *The conditional kernels $(P_t)_{t \geq 0}$ form a conservative strongly continuous Feller semigroup on a weighted uniformly continuous history space. For histories agreeing on the last L units,*

$$|P_t F(h) - P_t F(\tilde{h})| \leq C \|F\|_{\text{dyn}} (e^{-\gamma L} + e^{-\gamma t})$$

after the singularity-shield error is included. The future law is nondegenerate unless the prepared phase itself is deterministic.

Proof. The natural extension disintegrates a Gibbs path law into a one-sided g -kernel in the forward temporal orientation. Summable variations give uniqueness and exponential coupling of two futures whose finite pasts agree. The A3 singularity shield transfers the symbolic coupling to graph-completed billiard paths, while the exponential roof tail controls continuous-time renewal residuals. Conditioning successively and concatenating futures gives Chapman–Kolmogorov. The coupling estimate gives Feller continuity and strong continuity. Since the future coordinate was not retained in h , the conditional kernel is not a Dirac kernel. $\square$

1.2. The correctly oriented Poisson decomposition. Let g be centered and dynamically Hölder, and write $Pg(h) = \int g(h') P_1(h, dh')$. Put

$$h_g = \sum_{n \geq 1} P^n g, \quad m_g(h_0, h_1) = g(h_1) + h_g(h_1) - Ph_g(h_0).$$

Lemma 1.2 (Conditional martingale–coboundary decomposition). *The series for h_g converges in the weighted Hölder norm and*

$$\mathbb{E}[m_g(H_k, H_{k+1}) \mid \mathcal{F}_k] = 0.$$

Moreover

$$\sum_{j=1}^n g(H_j) = \sum_{j=0}^{n-1} m_g(H_j, H_{j+1}) + Ph_g(H_0) - Ph_g(H_n).$$

Proof. The Feller coupling estimate implies $\|P^n g\| \leq Ce^{-\gamma n} \|g\|$, so the Poisson series converges. By the Markov property,

$$\mathbb{E}[g(H_{k+1}) + h_g(H_{k+1}) \mid \mathcal{F}_k] = P(g + h_g)(H_k) = Ph_g(H_k),$$

because $(I - P)h_g = Pg$. This proves conditional centering. Summing the definition and using the Poisson identity gives the coboundary formula. The raw centered observable is never called a martingale difference. $\square$

Theorem 1.3 (Uniform quenched enhanced invariance principle). *For every compact regular family of prepared phases and every vector of weighted dynamically Hölder observables, the conditional laws given $H_0 = h$ of the diffusively rescaled sums and their canonical iterated integrals converge, locally uniformly in h , to a Brownian rough path. Its covariance and area drift are the predictable bracket and antisymmetric coboundary limits of Lemma 1.2.*

Proof. The martingale increments have a uniform $2 + \delta$ moment under the conditional kernels, by the roof and singularity exponential moments. Conditional Lindeberg and bracket convergence follow from Feller coupling and the ergodic theorem uniformly on compact history sets. The martingale rough functional CLT gives the first two levels. The coboundary is negligible at the first level and contributes the explicit antisymmetric area drift at the second. Truncation at Young returns and the A3 exponential shield remove the compact-history restriction. This proves stable, hence quenched, convergence. $\square$

1.3. The correct nonlinear history semigroup. For a bounded potential Ψ , let

$$T_t^\Psi F(h) = \mathbb{E}_h \left[e^{\int_0^t \Psi(H_s) ds} F(H_t) \right].$$

Let $e^{\lambda_\Psi t}$, $h_\Psi > 0$, and ℓ_Ψ be the simple leading Feynman–Kac eigen-data on the regular source chart.

Theorem 1.4 (Eigenfunction-normalized nonlinear semigroup). *The Doob kernels*

$$\tilde{P}_t^\Psi F = e^{-\lambda_\Psi t} h_\Psi^{-1} T_t^\Psi (h_\Psi F)$$

are conservative and satisfy the exact semigroup law. Consequently

$$\mathcal{U}_t^\Psi F = \log \tilde{P}_t^\Psi (e^F)$$

is a cash-additive nonlinear semigroup. The unnormalized logarithm $\log T_t^\Psi (e^F)$ also composes; the state-dependent quotient $\log T_t^\Psi e^F - \log T_t^\Psi 1$ is not used as a semigroup unless it is expressed through the displayed Doob transform.

Proof. The Feynman–Kac operators compose by the Markov property and satisfy $T_t^\Psi h_\Psi = e^{\lambda_\Psi t} h_\Psi$. Substitution therefore gives $\tilde{P}_{t+s}^\Psi = \tilde{P}_t^\Psi \tilde{P}_s^\Psi$ and $\tilde{P}_t^\Psi 1 = 1$. Exponentiating and taking logarithms gives the nonlinear tower and cash additivity. $\square$

1.4. Exact compressed-resolvent memory. Let L_Ψ be the generator of the Doob phase on its common Hilbert realization, P a finite-rank orthogonal resolved projection, and $Q = I - P$. Put

$$C(z) = P(z - L_\Psi)^{-1} P, \quad \widehat{K}(z) = zP - PL_\Psi P - C(z)^{-1}.$$

Theorem 1.5 (Full-Q dilation and causal memory). *On $P\mathcal{H} \oplus Q\mathcal{H}$, the fixed operator*

$$\mathbb{L}_\Psi = \begin{pmatrix} PL_\Psi P & PL_\Psi Q \\ QL_\Psi P & QL_\Psi Q \end{pmatrix}$$

has compressed resolvent $C(z)$. The resolved correlation satisfies

$$C'(t) = PL_\Psi P C(t) + (K * C)(t), \quad C(0) = P,$$

with the displayed sign convention. After adjoining the finite principal parts of $C(z)^{-1}$ as descriptor modes while retaining the full Q -space, the residual kernel is exponentially integrable on the spectral phase strip.

Proof. The Schur complement of $z - \mathbb{L}_\Psi$ is

$$zP - PL_\Psi P - PL_\Psi Q(z - QL_\Psi Q)^{-1}QL_\Psi P = C(z)^{-1}.$$

Rearranging gives the stated $\widehat{K}$. Inverse Laplace transformation yields the Volterra equation and its initial value. Meromorphic continuation from the A2 spectral packet has finitely many principal parts in every closed strip. A finite descriptor realization represents exactly those parts; it is coupled to, not substituted for, the unresolved Q -space. After their removal the contour moves to a negative line, giving exponential integrability of the residual memory. $\square$

Theorem 1.6 (History pressure and rough tangent). *The eigenfunction-normalized history semigroups converge under the diffusive scaling of Theorem 1.3 to the exponential semigroup of the Brownian rough differential equation with the proved covariance and area anomaly. The resolved memory coordinates converge jointly to the Volterra Gaussian lift generated by Theorem 1.5.*

Proof. Quenched rough convergence and uniform exponential moments pass bounded cylinder Feynman–Kac functionals to the rough limit. Analytic perturbation passes λ_Ψ and h_Ψ through the scaling, so the Doob normalization converges as well. The exponentially integrable residual memory is a continuous convolution functional, and the finite descriptor modes are finite-dimensional linear functionals. Joint convergence follows from the continuous mapping theorem. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs are used only in the acyclic order recorded in the round-ten dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks; they do not replace independent external mathematical review.

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